

SHETH L.U.J AND SIR M.V COLLEGE

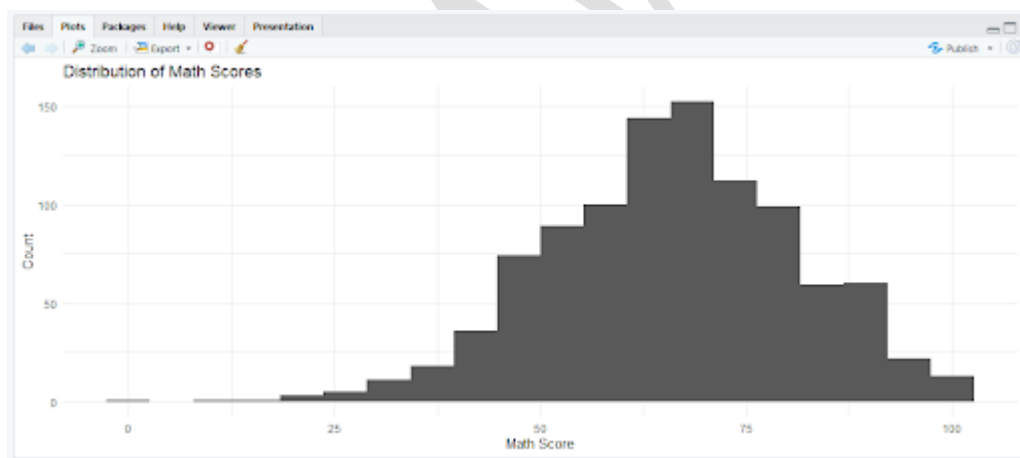
10 :-) Creating graphical reports using ,ggplot2 (R).

OUTPUT :-

```
> ggplot(data.frame(x = c("A", "B", "C")), aes(x)) +
+   geom_bar()
> library(ggplot2)
> library(dplyr)
>
> StudentsPerformance <- read.csv("C:/users/mvlu/downloads/StudentsPerformance.csv")
>
> str(StudentsPerformance)
'data.frame':   1000 obs. of  8 variables:
 $ gender      : chr  "female" "female" "female" "male" ...
 $ race.ethnicity : chr  "group B" "group C" "group B" "group A" ...
 $ parental.level.of.education: chr  "bachelor's degree" "some college" "master's degree" "associate's degree" ...
 $ lunch       : chr  "standard" "standard" "standard" "free/reduced" ...
 $ test.preparation.course : chr  "none" "completed" "none" "none" ...
 $ math.score   : int   72 69 90 47 76 71 88 40 64 38 ...
 $ reading.score : int   72 90 95 57 78 83 95 43 64 60 ...
 $ writing.score : int   74 88 93 44 75 78 92 39 67 50 ...
> head(StudentsPerformance)
  gender race.ethnicity parental.level.of.education lunch test.preparation.course math.score reading.score
1 female      group B      bachelor's degree      standard              none           72           72
2 female      group C           some college      standard              completed        69           90
3 female      group B      master's degree      standard              none           90           95
4 male        group A      associate's degree free/reduced              none           47           57
5 male        group C           some college      standard              none           76           78
6 female      group B      associate's degree      standard              none           71           83
  writing.score
1           74
2           88
3           93
4           44
5           75
6           78
>
> ggplot(StudentsPerformance, aes(x = gender)) +
+   geom_bar() +
+   labs(
+     title = "Number of Students by Gender",
+     x = "Gender",
+     y = "Number of Students"
```

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```
Console Terminal Background Jobs
R - R 4.1.2 - ~/
+ x = "Parental Education",
+ y = "Number of Students"
+ ) +
+ theme_minimal() +
+ theme(axis.text.x = element_text(angle = 45, hjust = 1))
>
> ggplot(StudentsPerformance, aes(x = gender, y = math.score)) +
+ geom_boxplot() +
+ labs(
+   title = "Math Score Distribution by Gender",
+   x = "Gender",
+   y = "Math Score"
+ ) +
+ theme_minimal()
>
> ggplot(StudentsPerformance, aes(x = reading.score, y = writing.score)) +
+ geom_point(alpha = 0.7) +
+ labs(
+   title = "Reading Score vs Writing Score",
+   x = "Reading Score",
+   y = "Writing Score"
+ ) +
+ theme_minimal()
>
> ggplot(StudentsPerformance, aes(x = lunch)) +
+ geom_bar() +
+ labs(
+   title = "Number of Students by Lunch Type",
+   x = "Lunch Type",
+   y = "Number of Students"
+ ) +
+ theme_minimal()
>
> ggplot(StudentsPerformance, aes(x = math.score)) +
+ geom_histogram(bins = 20) +
+ labs(
+   title = "Distribution of Math Scores",
+   x = "Math Score",
+   y = "Count"
+ ) +
+ theme_minimal()
> |
```



11 :-) Generating histograms and box plots using ggplot2 (R).

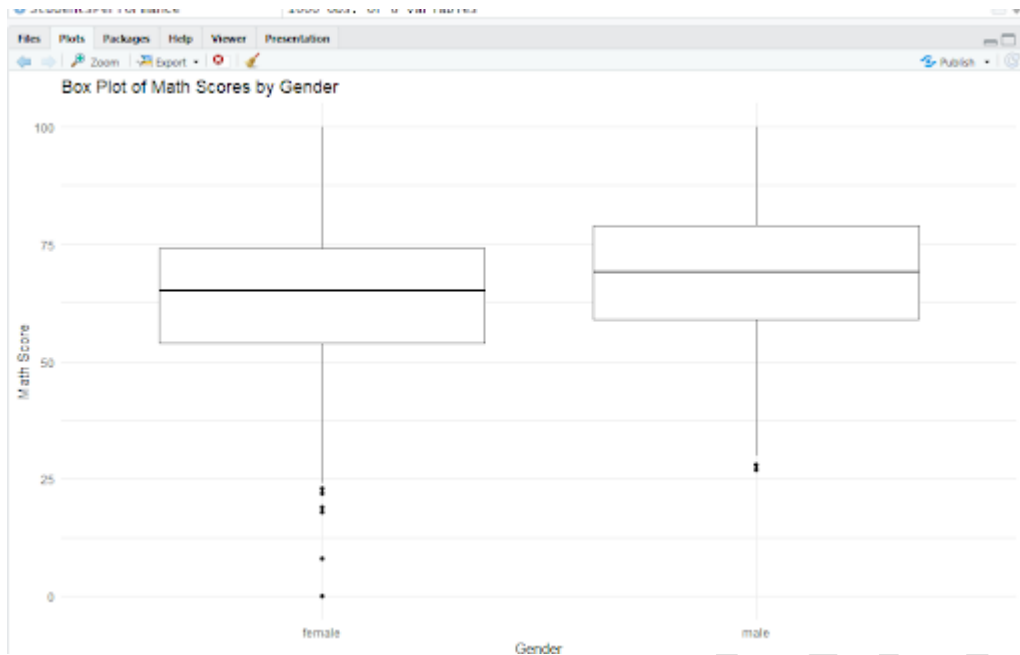
OUTPUT :-

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```
> StudentsPerformance <- read.csv("C:/Users/mvlui/Downloads/StudentsPerformance.csv")
> head(StudentsPerformance)
  gender race.ethnicity parental.level.of.education
1 female      group B      bachelor's degree
2 female      group C      some college
3 female      group B      master's degree
4 male        group A      associate's degree
5 male        group C      some college
6 female      group B      associate's degree
  lunch test.preparation.course math.score
1 standard      none           72
2 standard      completed      69
3 standard      none           90
4 free/reduced  none           47
5 standard      none           76
6 standard      none           71
  reading.score writing.score
1 72 74
2 90 88
3 95 93
4 57 44
5 78 75
6 83 78
> str(StudentsPerformance)
'data.frame': 1000 obs. of  8 variables:
 $ gender      : chr  "female" "female" "female" "male" ...
 $ race.ethnicity: chr  "group B" "group C" "group B" "group A" ...
 $ parental.level.of.education: chr  "bachelor's degree" "some college" "master's degree" "associate's degree" ...
 $ lunch       : chr  "standard" "standard" "standard" "free/reduced" ...
 $ test.preparation.course : chr  "none" "completed" "none" "none" ...
 $ math.score  : int  72 69 90 47 76 71 88 40 64 38 ...
 $ reading.score : int  72 90 95 57 78 83 95 43 64 60 ...
 $ writing.score : int  74 88 91 44 75 78 92 39 67 50 ...
> ggplot(StudentsPerformance, aes(x = math.score)) +
+   geom_histogram(binwidth = 5) +
+   labs(
+     title = "Histogram of Math Scores",
+     x = "Math Score",
+     y = "Frequency"
+   )
> ggplot(StudentsPerformance, aes(x = math.score)) +
+   geom_histogram(binwidth = 5) +
+   facet_wrap(~ gender) +
+   labs(
+     title = "Histogram of Math Scores by Gender",
+     x = "Math Score",
+     y = "Frequency"
+   ) +
+   theme_minimal()
> ggplot(StudentsPerformance, aes(y = math.score)) +
+   geom_boxplot() +
+   labs(
+     title = "Box Plot of Math Scores",
+     y = "Math Score"
+   ) +
+   theme_minimal()
> ggplot(StudentsPerformance, aes(x = gender, y = math.score)) +
+   geom_boxplot() +
+   labs(
+     title = "Box Plot of Math Scores by Gender",
+     x = "Gender",
+     y = "Math Score"
+   ) +
+   theme_minimal()
> |
```

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12 :- Generating correlation matrices using cor() (R).

OUTPUT :-

```
> StudentsPerformance <- read.csv("C:/Users/mvlu/Downloads/StudentsPerformance.csv")
> 
> head(StudentsPerformance)
  gender race,ethnicity parental.level.of.education lunch test.preparation.course
1 female      group B      bachelor's degree    standard                    none
2 female      group C      some college    standard                    completed
3 female      group B      master's degree    standard                    none
4 male        group A      associate's degree free/reduced    none
5 male        group C      some college    standard                    none
6 female      group B      associate's degree    standard                    none
  math.score reading.score writing.score
1         72          72          74
2         69          90          88
3         90          95          93
4         47          57          44
5         76          78          75
6         71          83          78
> str(StudentsPerformance)
'data.frame':   1000 obs. of  8 variables:
 $ gender      : chr  "female" "female" "female" "male" ...
 $ race,ethnicity: chr  "group B" "group C" "group B" "group A" ...
 $ parental.level.of.education: chr  "bachelor's degree" "some college" "master's degree" "associate's degree" ...
 $ lunch       : chr  "standard" "standard" "standard" "free/reduced" ...
 $ test.preparation.course : chr  "none" "completed" "none" "none" ...
 $ math.score  : int  72 69 90 47 76 71 88 40 64 38 ...
 $ reading.score: int  72 90 95 57 78 83 95 43 64 60 ...
 $ writing.score: int  74 88 93 44 75 78 92 39 67 50 ...
> 
> numeric_data <- StudentsPerformance[sapply(StudentsPerformance, is.numeric)]
> 
> head(numeric_data)
  math.score reading.score writing.score
1         72          72          74
2         69          90          88
3         90          95          93
4         47          57          44
5         76          78          75
6         71          83          78
> correlation_matrix <- cor(numeric_data, use = "complete.obs")
```

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